

NCERT Class 10 Science - Chapter 12: Magnetic Effects of Electric Current

12.1 Magnetic Field & Field Lines

Properties of Magnetic Field Lines: Emerge from North pole and merge at South pole outside magnet. Continuous closed loops. Field lines never intersect each other.

12.2 Right-Hand Thumb Rule

Right-Hand Thumb Rule: Hold straight conductor in right hand with thumb pointing in direction of current I . Curled fingers show direction of magnetic field lines B . Used for straight wire and solenoid.

12.3 Fleming's Left-Hand Rule

Fleming's Left-Hand Rule for Force: Stretch thumb, forefinger, and middle finger of LEFT hand perpendicular to each other. Forefinger = Field (B), Middle finger = Current (I), Thumb = Force / Motion (F). Predicts mechanical force on conductor inside magnetic field.

12.5 Electromagnetic Induction & Fleming's Right-Hand Rule

Electromagnetic Induction: Production of induced current due to changing magnetic flux. Fleming's Right-Hand Rule predicts direction of induced current.